# Digital Signal Processing Laboratory

System Stability Analysis

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**Abstract**

This report documents the findings of Laboratory 4 in the Digital Signal Processing course. The experiments explore fundamental concepts of discrete-time linear time-invariant (LTI) systems. Specifically, we investigate the calculation and visualization of Z-transform poles and zeros, analyze the time-domain impulse response of finite impulse response (FIR) filters, evaluate the frequency response of a highpass filter in decibels, and assess system stability based on the pole locations relative to the unit circle.

## 1 Introduction

The Z-transform is a powerful mathematical tool used extensively in digital signal processing for the analysis and design of discrete-time systems. By mapping discrete-time signals into the complex frequency domain, it simplifies the evaluation of system characteristics such as frequency response and stability. In this laboratory, MATLAB is utilized to bridge the gap between theoretical DSP concepts and practical computational analysis.

## 2 Experiment 4-1: Z-Transform Poles and Zeros

The transfer function of a discrete-time LTI system is typically represented as a rational function $H(z) = \frac{B(z)}{A(z)}$, where $B(z)$ and $A(z)$ are polynomials in $z^{-1}$. The roots of the numerator polynomial $B(z)$ are the *zeros* of the system, and the roots of the denominator polynomial $A(z)$ are the *poles*.

In this experiment, we explore two methods for visualizing these critical points on the complex plane. First, we directly input the polynomial coefficients into the `zplane` function, which internally calculates the roots and plots them relative to the unit circle. Second, we use the `tf2zp` (transfer function to zero-pole) command to explicitly calculate the exact numerical coordinates of the zeros and poles before passing them to `zplane`. Understanding the geometric layout of these poles and zeros is essential, as their proximity to the unit circle dictates the frequency-selective behavior of the system.

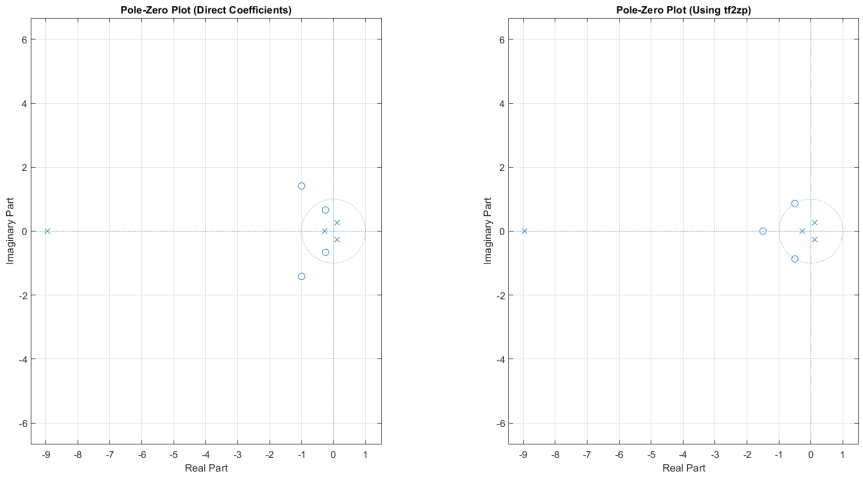



Figure 1: Pole-Zero plots using direct coefficients (left) and tf2zp function (right).

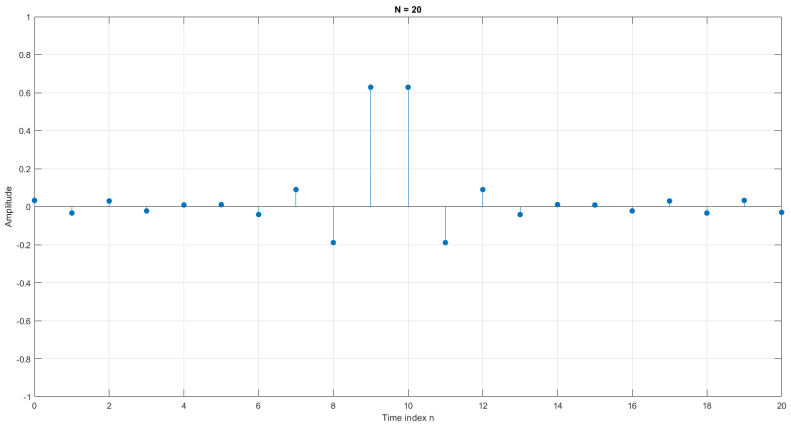



Figure 2: Stem plot of the discrete-time filter amplitude over time indices.

# 3 Experiment 4-2: System Properties and Filter Types

This experiment focuses on the time-domain characteristics of digital filters. An ideal lowpass filter has an impulse response that takes the shape of a sinc function, extending infinitely in both the positive and negative time directions. Because an infinite impulse response cannot be practically implemented in computational hardware, it must be truncated.

We define a truncated discrete-time signal governed by the parameter $f_c = 0.45$, which represents the normalized cutoff frequency. By generating this sequence over a finite number of samples (investigating lengths such as $N = 13$ and $N = 20$), we effectively design a Finite Impulse Response (FIR) filter using the windowing method (in this case, a rectangular window). Using the `stem` plot, we visualize the amplitude of the impulse response at discrete time indices. The results demonstrate how the length of the filter $N$ affects the time-domain representation, which directly correlates to the steepness of the filter's roll-off in the frequency domain.

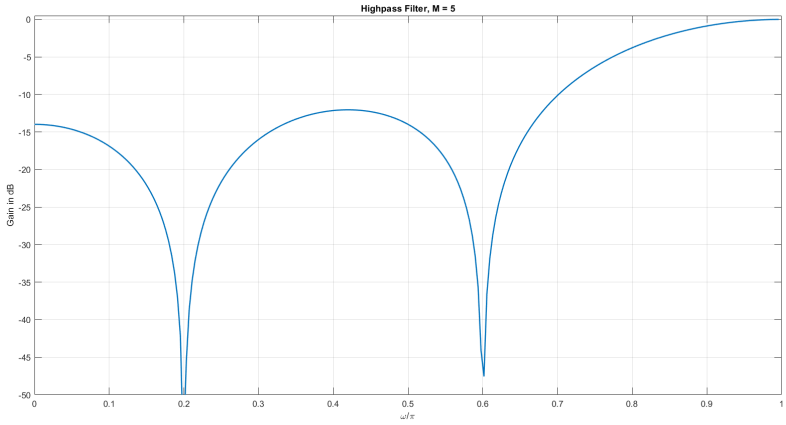



Figure 3: Frequency response gain (in dB) of the $M = 5$ highpass filter.

# 4 Experiment 4-3: Gain of a Highpass Filter

Evaluating a system's frequency response is a cornerstone of filter design. In this section, we analyze a highpass filter configuration defined by the parameter $M = 5$. The filter's coefficients are designed with alternating signs to suppress low frequencies and pass high frequencies.

To analyze its behavior, we evaluate the system's frequency response $H(e^{j\omega})$ over the normalized frequency range $0 \leq \omega \leq \pi$. The magnitude of the response is converted to the logarithmic decibel scale using the relation $20 \log_{10} |H(e^{j\omega})|$. Plotting the gain against the normalized frequency reveals the filter's spectral characteristics. We specifically identify the $-3$ dB point (the half-power frequency), which occurs at approximately $0.81\pi$ rad/sample. The plot clearly illustrates the passband, the transition band, and the attenuation in the stopband, confirming the system's highpass nature.

# 5 Experiment 4-4: System Stability Analysis

The stability of a discrete-time LTI system is strictly determined by the locations of its poles in the complex Z-plane. According to the Bounded-Input Bounded-Output (BIBO) stability criterion, a causal LTI system is stable if and only if all of its poles lie strictly inside the unit circle ($|z| < 1$).

We test this criterion by plotting the zero-pole diagrams for two closely related systems:

- **System 1 ($H_1(z)$):** Defined by the denominator coefficients $[1, -1.848, 0.85]$. The roots of this polynomial yield poles that are located just inside the unit circle (maximum magnitude approximately 0.9989). Because all poles satisfy $|z| < 1$, System 1 is determined to be strictly **stable**.

- **System 2 ($H_2(z)$):** Defined by a slightly perturbed set of denominator coefficients $[1, -1.851, 0.85]$. Upon calculating the roots, we observe that one of the poles has a magnitude greater than 1 (approximately 1.0005), placing it outside the unit circle. Consequently, System 2 is **unstable**.

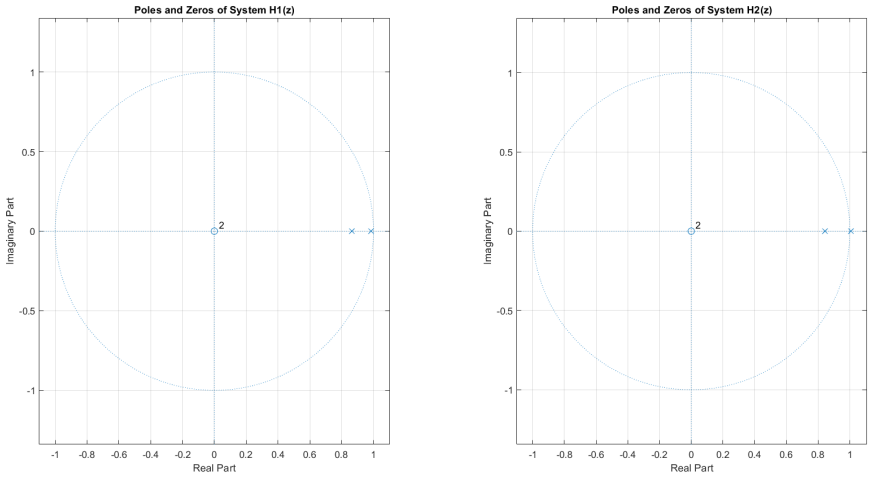



Figure 4: Pole-zero diagrams for System 1 (stable) and System 2 (unstable).

This experiment highlights the extreme sensitivity of system stability to minor variations in transfer function coefficients, emphasizing the necessity of high-precision arithmetic in recursive (IIR) digital filter implementations.

# 6 Conclusion

Through these four experiments, the theoretical foundations of the Z-transform and digital filtering were successfully verified via MATLAB simulations. We demonstrated how pole-zero placements dictate frequency response and rigorously applied the unit circle criterion to determine system stability. These principles are fundamental for the design and analysis of robust digital signal processing systems.